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Economic Evaluation

Methods to Include Environmental Impacts in a Cost-Effectiveness Analysis: A Case Study of Surveillance Imaging After Treatment for Stage III Melanoma

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ABSTRACT

Objectives: To demonstrate methods to include carbon emissions in a cost-effectiveness analysis of different positron emission tomography and computed tomography surveillance imaging schedules for detecting metastasis in patients treated for stage III melanoma.

Methods: Decision-analytic modeling was used to compare 12-monthly imaging with no routinely scheduled imaging. The health outcome was the proportion of patients without a false-negative result for a distant metastasis. Health system costs were included in 2023 Australian dollars. A 5% discount per year was applied. Carbon emissions were obtained from existing studies. Four methods were demonstrated: (1) reporting emissions in parallel, (2) the incremental carbon footprint cost ratio, (3) the incremental carbon footprint effectiveness ratio, and (4) integrating emissions as a cost in the incremental cost-effectiveness ratio (ICER).

Results: In the base case, 12-monthly imaging was more expensive and more effective than no scheduled imaging, with an ICER of $71 757 per false-negative avoided. Carbon emissions were higher for 12-monthly imaging (1497 kg CO_2-e) than no routine imaging (1037 kg CO_2-e). The incremental carbon footprint cost ratio was 0.2 kg CO_2-e per additional dollar spent. The incremental carbon footprint effectiveness ratio was 17 331 kg CO_2-e per false-negative avoided. Valuing emissions using New South Wales carbon values increased the ICER modestly to $73 919 per false-negative avoided. Using the mortality-accounting social cost of carbon increased the ICER to $107 297 per false-negative avoided.

Conclusions: Parallel methods are difficult to interpret without agreed value thresholds. Integrated methods meaningfully altered the results of the base-case analysis when higher monetary values of carbon emissions were used.

Keywords: carbon emissions, cost-effectiveness analysis, economic evaluation, environmental impact, health technology assessment, sustainable healthcare.

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Highlights

- This study explored methods to include carbon emissions in a modeled cost-effectiveness analysis.
- We demonstrated parallel and integrated methods to include carbon emissions in a decision-analytic model and found that these meaningfully changed the results of the base-case cost-effectiveness analysis when integrated as a cost using a high carbon value.
- The benefit of including carbon emissions in health economic evaluation may be in making impacts explicit and not in altering recommendations or decisions.

Introduction

Health economic evaluations inform decisions about the allocation of healthcare resources by comparing the costs and consequences of alternative healthcare interventions.[1] Common economic evaluation techniques include cost-effectiveness analysis (CEA), cost-utility analysis, cost-minimization analysis (CMA), and cost-benefit analysis. These are typically undertaken from the health system perspective, considering economic costs to the healthcare system and associated health benefits arising from the use of one intervention over another. Broader perspectives, such as patient or societal perspectives, may include other costs such as patient out-of-pocket costs, productivity gains or losses, and costs to other nonhealth sectors. Regardless of the evaluation technique or perspective, environmental impacts have not traditionally been considered in health economic evaluations.

Healthcare interventions have large environmental impacts. These include carbon emissions that contribute to climate change and have negative health impacts.[2] There is international interest in determining whether environmental impacts, particularly carbon emissions, should be included in health economic evaluations, as part of broader efforts to improve the environmental sustainability of health systems.[3,4] Existing work suggests including carbon emissions in economic methods is feasible, and several studies have reviewed the methodological options for doing so.[5–8]

Much of the methodological literature focuses on the role of economic evaluations in Health Technology Assessment (HTA). This is driven by work from HTA agencies in Australia,[9] and elsewhere,[10,11] to understand how environmental information might be included in their processes. Despite recent work, there are few empirical studies demonstrating the practical application of these methods. A CMA from Australia included

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greenhouse gas (GHG) emissions and solid waste in an evaluation of single-use hospital gowns.[12] A CEA from the United Kingdom included GHG emissions in an evaluation of different obesity treatments.[13] No study has directly compared the application of different methods to include carbon emissions to the same case study.

The aim of this study was to demonstrate different methods to include environmental impacts in health economic evaluations. Specifically, we aimed to demonstrate parallel and integrated methods to include carbon emissions (CO_2-e) in a modeled CEA of different positron emission tomography and computed tomography (PET/CT) surveillance imaging schedules for detecting distant melanoma recurrence.

Methods

Overview of Methodology

This study used a decision-analytic model to explore the cost-effectiveness of different PET/CT schedules in avoiding false-negative test results for distant melanoma metastases. First, a CEA was performed that did not include environmental impacts (ie, the base case). Second, methods to include carbon emissions in the CEA were identified and described. Finally, 4 methods were demonstrated and compared with the results of the base-case CEA. A health economic analysis plan was not developed for this study.

The decision-analytic model was built using TreeAge Pro (Healthcare Version) 2025 (TreeAge Software, LLC: Williamstown, MA, USA). Additional data analyses were conducted using R version 4.4.3 and R Studio Version 2024.12.1+563 (Posit Software PBC: Boston, MA, USA). The study was reported according to the Consolidated Health Economic Evaluation Reporting Standards (CHEERS) 2022 Statement (Appendix Table 1).[14]

Case Study

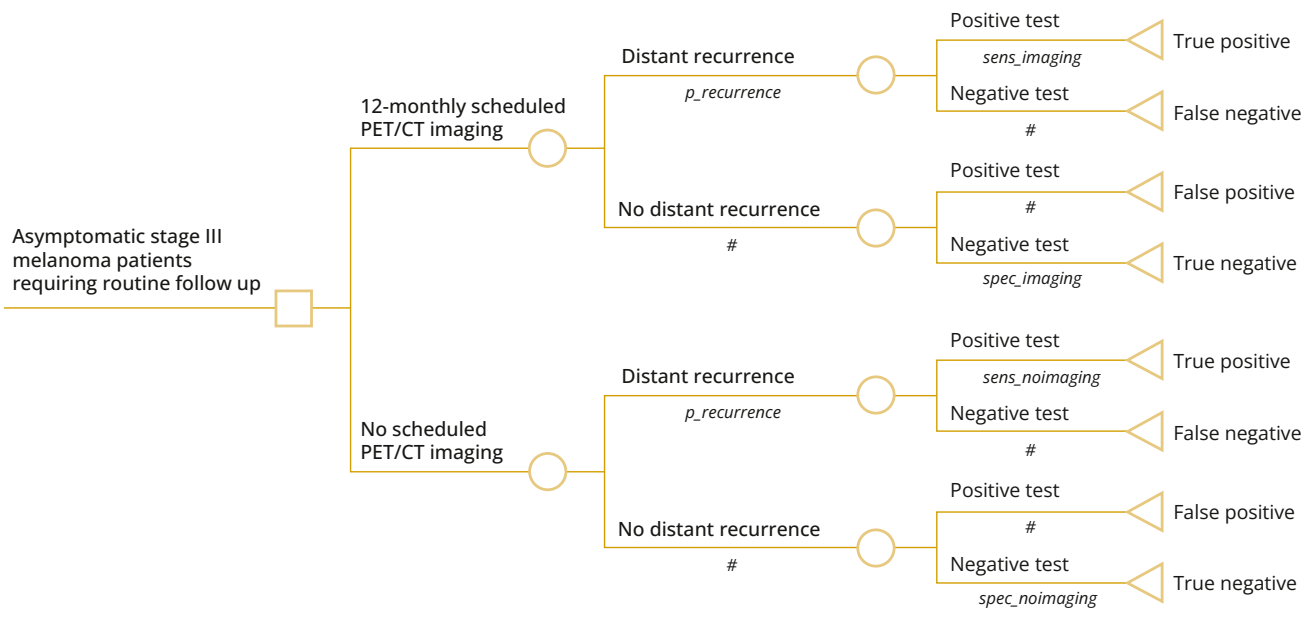
Australia has the highest incidence of cutaneous melanoma in the world. This study explored the surveillance of patients after surgical treatment for American Joint Committee on Cancer stage III melanoma including affected lymph nodes. These patients have a high chance of a distant metastasis and undergo frequent follow-up examinations with the aim of detecting and treating the recurrence early. Examinations may include the use of diagnostic imaging such as PET/CT. The benefit of these imaging tests has not been demonstrated, and recommendations vary on how frequently they should be done. Previous studies have explored this population in detail, including describing test performance,[15] cost and carbon impact,[16] and cost-effectiveness.[17]

Decision-Analytic Model

The analysis presented in this study assesses the relative cost-effectiveness of 12-monthly PET/CT imaging compared with no scheduled PET/CT imaging for avoiding false-negative test results (missed recurrence) during the surveillance of adults with stage III melanoma. The analysis relied on the structure of an existing decision model, details of which are available elsewhere.[17]

The decision node was the choice of 12-monthly or no scheduled PET/CT imaging. Distant recurrence (recurrence or no recurrence) and test result (positive or negative) were included as chance nodes. The 4 possible terminal nodes per decision were as follows: true positive, false positive, true negative and false negative. The structure of the model is provided in Figure 1. The analysis was conducted over a 5-year time horizon from the health system perspective. A 5% discount rate for costs and outcomes was applied per year. Probabilities were assigned to each branch of the model for the 5-year rate of recurrence and the sensitivity and specificity of each surveillance strategy, based on data from existing studies. Rate of recurrence was obtained from a recent randomized controlled trial (RCT) and was assumed to be

Figure 1. Structure of decision-analytic model to determine the cost-effectiveness of different surveillance imaging schedules for resected stage III melanoma patients to avoid a false-negative test result. Inputs for the standard cost-effectiveness analysis are shown. Model inputs are described in Table 1.



35% (COMBI-AD RCT intervention group)[18,19]; this represents chance of recurrence in a clinical population after complete resection of stage III melanoma receiving a combination of dabrafenib and trametinib. Clinical examination was assumed to have a sensitivity of 71% and a specificity of >99%.[20] PET/CT imaging was assumed to have a sensitivity of 79% and a specificity of 88%.[15] Model inputs are detailed in Table 1.[21-28]

The health outcome of the analysis was the proportion of patients avoiding a false negative. There is no empirical evidence that avoiding false negatives is associated with survival.[29] However, it is a plausible surrogate that can be used for clinical decisions. To achieve this, binary health payoffs (0 or 1) were assigned to each terminal node of the model so that a false negative was assigned a payoff of 0 and all other outcomes (ie, true positive, true negative, and false positive) were assigned a payoff of 1.

Costs in 2023 Australian dollars were included for surveillance imaging activities, clinic appointments, diagnostic confirmation of a positive test result, and treatment of recurrent disease. The cost of surveillance imaging for each schedule was obtained from an existing cohort study ($3835 for 12-monthly imaging and $1595 for no scheduled imaging).[16] The cost of treatment was assumed to be the 12-month cost of systemic treatment for stage III unresectable or stage IV melanoma and was obtained from an existing study.[21] The cost of confirmation of a positive test result was assumed to be the cost of a fine-needle aspiration biopsy for 12-monthly imaging and a fine-needle aspiration biopsy and whole-body PET/CT for no scheduled imaging and were obtained from the Australian Medicare Benefits Schedule. The cost of 1 clinic appointment was also included for all patients with a recurrence or positive test result. Costs were adjusted to 2023 AUD using the Total Health Price Index as appropriate.[30] Details are provided in Table 1.

Base-Case Cost-Effectiveness Analysis

A base-case CEA that did not include carbon emissions was performed to act as a reference to which demonstrated methods were compared. The primary outcome of the CEA was the incremental cost-effectiveness ratio (ICER), being the cost per false-negative avoided. Expected costs and outcomes were reported for each imaging schedule. The ICER was calculated using 12-monthly imaging as the intervention and no routinely scheduled imaging as the comparator. A 5% discount per year was applied to future costs and outcomes to account for the 5-year time horizon, consistent with recommendations from the Pharmaceutical Benefits Advisory Committee (Appendix Table 2).[31]

Table 1. Model inputs.

Input	Variable	Base-case analysis	Sensitivity analysis	Source
5-year rate of distant melanoma recurrence	*p_recurrence*	0.35	0.29-0.39	Dummer et al[18]; Long et al[19]
Sensitivity of PET/CT imaging	*sens_imag*	0.79	0.70-0.86	Turner et al[15]
Specificity of PET/CT imaging	*spec_imag*	0.88	0.86-0.90	Turner et al[15]
Sensitivity of clinical examination	*sens_noimag*	0.714	0.554-0.843	Machet et al[20]
Specificity of clinical examination	*spec_noimag*	0.996	0.992-0.998	Machet et al[20]
Cost of 12-monthly scheduled PET/CT imaging	*c_12month_imaging*	$3,835	3828-3842	Williams et al[16]
Cost of no scheduled PET/CT imaging	*c_no_imaging*	$1,595	1588-1601	Williams et al[16]
Cost of treatment for distant recurrence*	*c_treatment*	$133,339.51	$121,752.03-$145,507.53	Elliott et al[21]
Cost of FNAB†	*c_FNAB*	$231.15	-	MBS 55054 MBS 30075
Cost of PET/CT†	*c_PETCT*	$996.10	-	MBS 61505 MBS 61553
Cost of clinic visit†	*c_clinicvisit*	$68.90	-	MBS 116
Environmental impact of 12-monthly scheduled PET/CT imaging‡	*e_12month_imaging*	750 kg CO_2-e	749-755	Williams et al[16]
Environmental impact of no scheduled PET/CT imaging‡	*e_no_imaging*	250 kg CO_2-e	245.5-250.5	Williams et al[16]
Environmental impact of treatment for distant recurrence‡	*e_treatment*	2,117.3 kg CO_2-e	-	Malmberg et al[22]; Dokal et al[23]; Long et al[24]; Australian Institute of Health and Welfare[25]
Environmental impact of fine needle aspiration biopsy‡	*e_FNAB*	0.647 kg CO_2-e	-	McAlister et al[26]; McAlister et al[27]
Environmental impact of PET/CT‡	*e_PETCT*	115 kg CO_2-e	-	Williams et al[16]; McAlister et al[27]
Environmental impact of clinic visit‡,§	*e_clinicvisit*	13.9 kg CO_2-e	7.19-23.9	Williams et al[28]

FNAB indicates fine-needle aspiration biopsy; MBS, Medicare Benefits Schedule item number; PET/CT, positron emission tomography and computed tomography.
*Cost adjusted from 2018 AUD to 2023 AUD using the Total Health Price Index.
†Cost identified using MBS item numbers as of 1 July 2022.
‡Detailed assumptions and methods for the calculation of environmental impact are included in Appendix Table 3.
§Data obtained from the preliminary results of a currently unpublished life cycle assessment of a clinic appointment for a full skin examination by a dermatologist.

One-way sensitivity analyses were conducted to explore parameter uncertainty on the primary outcome. Each input was varied between its likely minimum and maximum values as described in Table 1. These values came from published confidence intervals for each input. A tornado diagram was used to explore which inputs had the most impact on the primary outcome (ICER). An additional sensitivity analysis was conducted using higher 5-year rates of recurrence (52% and 61%) based on data from a recent RCT (EORTIC RCT intervention and control groups)[19,32]; this represents the chance of recurrence in a clinical population after complete resection of stage III disease receiving ipilimumab (intervention) or placebo (control), respectively. An additional sensitivity analysis was conducted with a lower estimated 5-year rate of recurrence of 20% based on findings from the recent NADINA trial (neoadjuvant immunotherapy).[33]

Identifying Methods to Include Environmental Impacts

Methods to include environmental impacts in a CEA were identified from our scoping review on the topic.[5] Appropriate methods were either parallel or integrated in their approach and are summarized below.[5,6]

Parallel evaluations are those that analyze and present environmental impact data alongside metrics of established health economic methods, for example, alongside the total costs, total health outcomes, and ICER per unit of health benefit gained. These methods do not usually monetize environmental impacts or require changes to existing economic methods.[6] In this study we demonstrated 3 parallel methods. First, we reported total and incremental carbon emissions alongside the base-case results of the CEA (Method 1). Second, we reported an incremental carbon footprint effectiveness ratio (ICFER) alongside the base-case results of the CEA (Method 2).[13,34] Third, we reported an incremental carbon footprint cost ratio (ICFCR) alongside the base-case results of the CEA (Method 3).[13]

Integrated evaluations fully integrate environmental impacts in a way that allows clinical, financial, and environmental information to be synthesized into a single evaluation or ratio.[6] In this study we demonstrated an integrated method to include carbon emissions as a monetized cost in addition to healthcare costs in the CEA (Method 4).

Including Environmental Impacts in the Model

Carbon emissions were added to the model for each healthcare activity (ie, diagnostic imaging, confirmation tests, clinic visits, and treatment with immunotherapy) based on existing studies and preferably from process-based life cycle assessment (LCA). For parallel methods, environmental impacts were added to the model as a separate payoff. For the integrated method, environmental impacts were monetized and added to other health system costs. Environmental impact inputs are summarized in Table 1 and further details are provided in Appendix Table 3. Where an estimate was not available for an activity, assumptions were made based on similar activities and these are detailed in Appendix Table 3.

Demonstrating Methods

Method 1: Parallel results

The primary outcomes of this analysis were the ICER and incremental carbon emissions. Expected costs, outcomes, and carbon emissions were reported for each imaging schedule. The ICER was the same as in the base-case analysis. Incremental carbon emissions were calculated for 12-monthly imaging using no scheduled imaging as the comparator. Carbon emissions (in kg CO_2-e) were not discounted consistent with previous recommendations to ensure intergenerational equity (Appendix Table 2).[35]

Method 2: Incremental carbon footprint cost ratio

The ICFCR was calculated by dividing incremental carbon emissions by incremental healthcare costs.[13] The ICFCR described the relationship between costs and the environment as emissions expected per dollar spent on an intervention. The ICFCR may be useful to understand the carbon emissions associated with money spent from the health budget and whether this is consistent with any stated emissions reduction goals. We calculated the ICFCR for 12-monthly imaging using the no scheduled imaging strategy as the comparator. Expected environmental impacts, incremental environmental impacts, and the ICFCR were reported parallel to the results of the base-case analysis.

Method 3: Incremental carbon footprint effectiveness ratio

The ICFER, also called the carbon-effectiveness ratio, was calculated by dividing incremental carbon emissions by incremental health outcomes.[13,34] The ICFER described the trade-off between health outcomes and the environment as the emissions expected per health unit gained. We calculated an ICFER for 12-monthly imaging using the no scheduled imaging strategy as the comparator. Expected environmental impact, incremental environmental impact, and the ICFER were reported parallel to the results of base-case analysis.

Method 4: Integrated evaluation

We integrated carbon emissions into the CEA on the cost side of the ICER. There are 2 common ways to value carbon emissions: using either a damage cost or target-consistent approach.[36,37] A damage cost approach values carbon emissions as the monetary marginal damage to society of additional carbon emissions, such as the social cost of carbon. A target-consistent approach values emissions as the monetary marginal abatement cost of achieving defined emissions reduction targets.

In the primary analysis of this method a target-consistent approach was used, and emissions were valued in Australian dollars using values from the New South Wales Carbon Emissions in the Investment Framework ($130 per ton CO_2-e for FY25 in 2024 AUD).[37] This was adjusted to 2023 AUD using the Reserve Bank of Australia Inflation Calculator ($124.75 per ton CO_2-e). Expected carbon costs (in AUD) were reported separately and combined with healthcare costs. An ICER was calculated for the incremental total costs to avoid an additional false-negative test result. A 5% discount per year was applied to future costs and outcomes to account for the 5-year time horizon (Appendix Table 2). In this method carbon emissions were discounted as with any other cost, consistent with previous recommendations.[35]

A sensitivity analysis was conducted to explore the effect of using higher ($220.72 per ton CO_2-e) and lower ($86.37 per ton CO_2-e) carbon values from the New South Wales Government.[37] These reflect higher and lower than expected marginal abatement costs. An additional analysis was undertaken that used a social cost of carbon (SCC) of $328.88 (2023 AUD) per ton CO_2-e, adjusted from $185 (2020 USD) per ton CO_2-e.[38] Finally, an analysis was undertaken using a mortality-accounting SCC (mSCC). Estimates of SCC typically do not include the cost of mortality. The mSCC is a proposed SCC that accounts for future mortality, and in a hypothetical example was estimated to be at least $1373 (2021 USD) per ton CO_2-e.[39,40] This was adjusted to $2055.45 (2023 AUD) per ton.

Results

Base-Case Cost-Effectiveness Analysis

The total per person discounted costs at 5 years were $42 194 for 12-monthly imaging and $40 601 for no scheduled imaging. The discounted proportion of patients without a false-negative test result was 77% for 12-monthly imaging and 75% for no routinely scheduled imaging. Therefore, 12-monthly imaging was both more expensive (+$1593) and more effective (+2%) compared with no routine imaging. The ICER of 12-monthly imaging compared with no routinely scheduled imaging was $71 757 per false-negative avoided. Details are reported in Table 2.

The ICER was most sensitive to the sensitivity of PET/CT imaging, the sensitivity of clinical examination, and the probability of distant recurrence (Appendix Fig. 1). Sensitivity analyses exploring a range of 5-year recurrence found 12-monthly imaging became less cost-effective as the probability of recurrence decreased (Appendix Table 4).

Method 1: Report Parallel Results

Expected carbon emissions at 5 years were 1497 kg CO_2-e for 12-monthly imaging and 1037 kg CO_2-e for no scheduled imaging. Compared with no scheduled imaging, 12-monthly imaging had an incremental carbon impact of 461 kg CO_2-e. Details are reported in Table 2.

Method 2: Incremental Carbon Footprint Cost Ratio

The ICFCR of 12-monthly imaging compared with no scheduled imaging was 0.2 kg CO_2-e per additional dollar spent (Table 2).

Method 3: Incremental Carbon Footprint Effectiveness Ratio

The ICFER of 12-monthly imaging compared with no scheduled imaging was 17 331 kg CO_2-e per false-negative avoided.

Method 4: Integrated Evaluation as a Cost

Expected discounted carbon emissions cost $156 for 12-monthly imaging and $108 for no scheduled imaging when valued in Australian Dollars. Total discounted costs were therefore expected to be $42 349 for 12-monthly imaging and $40 708 for no scheduled imaging, of which carbon emissions accounted for 0.3% in both schedules. The expected discounted proportion of patients without a false-negative test result was 77% for 12-monthly imaging and 75% for no scheduled imaging. Therefore, 12-monthly imaging was both more expensive (+$1641) and more effective (+2%) compared with no routine imaging. The ICER of 12-monthly imaging compared with no scheduled imaging was $73 919 per false-negative avoided. Details are reported in Table 2.

Higher and lower NSW carbon values had minimal impact on the calculation of expected costs or the ICER (Appendix Table 5). The use of a mSCC had a substantial impact on the ICER and increased it to $107 297 per false-negative avoided. Compared with the analysis using NSW carbon values this increased the proportion of total costs attributable to carbon emissions from 0.3% to 6% for 12-monthly imaging and from 0.3% to 5% for no routine imaging.

Discussion

The results of the base-case analysis suggest 12-monthly PET/CT imaging was both more expensive and more effective than no scheduled PET/CT imaging, with an ICER of $71 757 per false-negative avoided (calculated using discounted result). The cost-effectiveness result was largely driven by the sensitivity of PET/CT imaging, the sensitivity of clinical examination, and the probability of distant recurrence. Carbon emissions were higher for 12-monthly imaging than for no routine imaging. Parallel methods describe this result but may be more difficult to interpret without agreed value thresholds. Integrated methods meaningfully altered the results of the base-case cost-effectiveness analysis when higher monetary values of carbon emissions

Table 2. Primary outcome of cost-effectiveness analysis.

		12-monthly	No routine*
Base-case cost-effectiveness analysis	Effect[†]	0.7721	0.7499
	Incremental effect[†]	0.0222	-
	Healthcare cost, AUD	$42 194	$40 601
	Incremental cost, AUD	$1593	-
	ICER, cost per unit of effect[†]	$71 757	-
Method 1: Report parallel results	Carbon emissions, kg CO_2-e	1497 kg	1037 kg
	Incremental carbon emissions	461 kg	-
Method 2: ICFCR	ICFCR, kg CO_2-e per AUD	0.2	-
Method 3: ICFER	ICFER, kg CO_2-e per effect[†]	17 331	-
Method 4: Integrated evaluation as cost	Healthcare cost, AUD	$42 194	$40 601
	Carbon emissions, AUD	$156	$108
	Total costs, AUD[‡]	$42 349	$40 708
	Incremental cost, AUD	$1641	-
	ICER, cost per unit of effect[†]	$73 919	-

Method 1 (parallel) summarizes the incremental carbon emissions of each schedule. Method 2 (parallel) reports the incremental carbon footprint cost ratio, being emissions expected per dollar spent. Method 3 (parallel) reports the incremental carbon footprint effectiveness ratio, being emissions expected per unit of health gained. Method 4 (integrated) integrates carbon emissions as a cost into the incremental cost-effectiveness ratio.
AUD indicates Australian dollars; ICER, incremental cost-effectiveness ratio; ICFCR, incremental carbon footprint cost ratio.
*Used as comparator for incremental calculation.
[†]Proportion of outcomes that are not false negatives.
[‡]Total cost is the sum of health care costs and environmental costs.

were used. We contribute to a large and growing body of work exploring methods to include environmental impacts in economic evaluations and HTA.[5,7,41-44]

Parallel methods described higher environmental impacts of 12-monthly imaging compared with no scheduled imaging (Method 1). The ICFCR (Method 2) and ICFER (Method 3) were both positive, describing the carbon emissions expected per additional dollar spent, and per false-negative avoided, respectively. The ICFER describes the potential trade-offs between carbon emissions and health benefits, whereas the ICFCR presents the relationship between an intervention's cost and its carbon emissions. It remains unclear how this information should be interpreted during reimbursement decision making.

Preliminary work from the United Kingdom suggests that the public may support HTA recommendations that approve technologies with desirable environmental outcomes but undesirable economic outcomes, so long as certain (undefined) thresholds are not passed.[45] However, the scenario where an intervention is cost-effective but is not approved because of higher carbon emissions is more controversial and has not been considered. Australian perspectives on these trade-offs have not yet been explored. The usefulness of the ICFCR and ICFER are limited until such perspectives have been explored and potential thresholds, such as those implied for the ICER, can be established. The ICFER may be more intuitive to decision makers used to interpreting ICERs because they describe similar trade-offs between a cost (either carbon emissions or dollars) and a health outcome. However, the use of multiple ratios may complicate decision making and the interpretation of overall value for money.

For our study, the results of the integrated method were consistent with the base-case CEA. When valued in dollars, environmental impacts accounted for a small proportion of the total cost of the intervention and comparator, respectively. However, this is sensitive to the valuation of carbon emissions. Across the range of monetary values used by the NSW Government for carbon emissions, the ICER did not meaningfully change ($73 243 to $77 432 per false-negative avoided for carbon values between $86.37 and $328.88 per ton CO_2-e). When the mortality-accounting social cost of carbon was used the ICER increased to $107 297. Although previous research suggests integrating carbon emissions as a cost does not often affect the results of the study,[12,46,47] we find that doing so can be impactful when high carbon values such as the mSCC are used.

The carbon values used in this study are underpinned by different theoretical and methodical approaches that should be explored. The preference to use target-consistent or damage cost carbon values is debated, and future work is needed to explore the relevance of these debates to health economic evaluation. This is particularly true if integrated evaluations emerge as the preferred method for including environmental impacts. Studies suggest that SCCs used in decision making are too low.[38,48] For example, 2 recent studies from the United States found that a SCC of $186 USD[38] or $284 2020 USD[48] per ton CO_2-e should be used instead of the previously used value of $51 USD per ton CO_2-e. It is possible that the NSW Government carbon value is also underestimated, though this was considered by our sensitivity analysis exploring higher carbon values. The mSCC expands on SCC estimates to include the mortality cost of carbon.[39,40] This appears to be impactful and should be explored further for use in economic evaluations.

Another important consideration is the perspective of the payer. Reporting environmental impacts may only be relevant if the payer is interested in reducing environmental impacts. This may be relevant for Governments with legislated GHG emission targets, or specific healthcare environmental sustainability goals. However, these methods may be less useful for payers without this motivation, such as for insurance companies (without regulatory motivations).

There are different ways to quantify the environmental impact of interventions in a health economic evaluation. In this study, we identified carbon emissions for each healthcare resource from existing literature. This is similar to the approach taken by de Preux and Rizmie, who rely on an existing published estimate of the carbon footprint of their treatment options to conduct an integrated CEA,[47] and Pugliesi et al, who rely on emissions factors for each resource identified in their microcosting study of robot-assisted total knee arthroplasty.[46] Ideally, these estimates would come from high-quality process-based life-cycle assessments. However, these are unlikely to be available for all healthcare resources for which an estimate is needed, and assumptions should be made based on the data that are available, as we have done in this study. Future studies should interrogate sources of environmental impact data as these become available, such as The Lancet MedZero,[49] ECOVAMED,[50] and HealthcareLCA.[51]

A slightly different approach was taken by Dunbar et al. who reported a simplified life-cycle assessment within their cost-minimization analysis of reusable gowns.[12] Although this option may be preferred in some instances, we suggest that it is unlikely to be realistic for most health economic evaluations, given the specialist expertise and software required to conduct a LCA.

Kindred et al use emissions intensity data to estimate GHG emissions directly from their estimate of the cost of each treatment being studied.[13] Although conversions from monetary costs are simple, they cannot accurately estimate the environmental impact of healthcare interventions at the granularity required in a health economic evaluation. Although this may be tempting in situations in which existing estimates of carbon emissions do not exist, we suggest it be avoided if possible. This is because estimates of carbon emissions (in kg CO_2-e) will likely be higher if they are generated using methods that rely on cost information and lower if they are generated by bottom-up methods, as was used in this study and in those relying on process-based LCA or emissions factors for noncost resources.[52] This is important because cost-based methods may show larger differences in carbon emissions than are actually taking place, and this might incorrectly impact the CEA and any related decisions.

An approach unexplored in this study integrates environmental impacts in health units on the benefits side of the ICER. Ducrot et al. recently proposed a conversion factor of 1.18×10^{-5} quality-adjusted life-years per kg CO_2-e ($0.44 \times 10^{-5} - 4.83 \times 10^{-5}$).[39] Although the conversion of carbon emissions to health units is associated with significant uncertainty, it is a potentially useful option and should be considered by future studies. It was not included in the current case study because quality or disability adjusted survival (ie, QALYs or DALYs) were not used as the health outcome. The incremental carbon emissions of 461 kg CO_2-e reported in this study (Table 2) equates to a decrement of 2.0 days of life in full health (ie, 4.5×10^{-3} QALY) using the proposed conversion factor.

It is possible that including carbon emissions in a CEA may often not meaningly affect the interpretation of the analysis. Further work is needed to consider this in a variety of case studies to identify situations where emissions are most likely to affect funding decisions.[35,53,54] The broader value of including carbon emissions in an economic evaluation may not be in changing the decision made but in making emissions of healthcare explicit. Methods to include carbon emissions in HTA processes, instead of CEA, may be more useful in altering decisions. For example, methods such as multicriteria decision analysis (see, for example, a recent study by Kingma et al),[55] or deliberative

decision processes, may give greater explicit or implicit weight to carbon emissions related to cost-effectiveness. Future work could explore this possibility. For carbon emissions and other environmental impacts to be routinely included in CEA work is needed to establish guidelines for economic analysis plans and for reporting.[8] Work is ongoing to explore an extension to the CHEERS checklist to this effect.[56]

This study has several important limitations. First, the model structure is limited to a simplified 5-year patient pathway. This is consistent with the stated limitations of the existing model and, additionally, does not account for radiation burden (of PET/CT imaging) or lifetime costs and health outcomes. Second, the choice of health outcome as the proportion of false-negatives avoided is an unvalidated surrogate outcome, with the (unproven) assumption that earlier detection and treatment of metastatic melanomas improve health outcomes. This limits the interpretation of our CEA because the willingness-to-pay to avoid a false negative (ie, missed cancer) has not been established. Third, we do not demonstrate all possible methods to include environmental impacts nor any other methods to include impacts in HTA, such as multicriteria decision analysis or deliberative decision processes. Fourth, because of limitations in existing life-cycle assessment data, some inputs for environmental impacts are transposed from estimates for similar activities (Appendix Table 3). This example is also highly specific, and although our findings are consistent with existing work to include carbon emissions in a CEA, future work should explore other case studies, particularly in cases which the carbon emissions of the intervention are known or expected to be large and of particular concern.

Conclusions

This study compared and demonstrated parallel and integrated methods to include carbon emissions in a CEA. For this case example, we found that integrated methods were likely to change the interpretation of the base-case cost-effectiveness result that did not include carbon emissions when high carbon values were used. Parallel methods were difficult to interpret without understanding potential thresholds for their use. The benefit of including carbon emissions in economic evaluation may be in making environmental impacts explicit.

Author Disclosures

Author disclosure forms can be accessed below in the Supplemental Material section.

Supplemental Material

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REFERENCES

1. Drummond MF, Sculpher MJ, Claxton K, Stoddart GL, Torrance GW. *Methods for the Economic Evaluation of Health Care Programmes*. 4th ed. Oxford: Oxford University Press; 2015.
2. Romanello M, Walawender M, Hsu SC, et al. The 2024 report of the Lancet Countdown on health and climate change: facing record-breaking threats from delayed action. *The Lancet*. 2024;404(10465):1847–1896.
3. MacNeill AJ, McGain F, Sherman JD. Planetary health care: a framework for sustainable health systems. *Lancet Planet Health*. 2021;5(2):e66–e68.
4. Mortimer F. The sustainable physician. *Clin Med (Lond)*. 2010;10(2):110–111.
5. Williams JTW, Bell KJL, Morton RL, Dieng M. Methods to include environmental impacts in health economic evaluations and health technology assessments: a scoping review. *Value Health*. 2024;27(6):794–804.
6. Toolan M, Walpole S, Shah K, et al. Environmental impact assessment in health technology assessment: principles, approaches, and challenges. *Int J Technol Assess Health Care*. 2023;39(1):e13.
7. Desterbecq C, Tubeuf S. Inclusion of environmental spillovers in applied economic evaluations of healthcare products. *Value Health*. 2023;26(8):1270–1281.
8. Oldenburg J, Lange O, Keil M, et al. Points to consider for incorporating climate impacts into health economic evaluation. *Eur J Health Econ*. 2025. Forthcoming.
9. Australian Government Department of Health and Aged Care. Health Technology Assessment Policy and Methods Review – Final Report. https://www.health.gov.au/resources/publications/health-technology-assessment-policy-and-methods-review-final-report; 2024. Accessed October 16, 2024.
10. National Institute for Health and Care Excellence (NICE). HTA Lab projects. National Institute for Health and Care Excellence (NICE). https://www.nice.org.uk/about/what-we-do/our-research-work/hta-lab-projects. Accessed March 31, 2025.
11. Canada's Drug Agency. Ahead of the curve: shaping future-ready health systems - 2022-2025 strategic plan. Canada's Drug Agency. https://www.cda-amc.ca/2022-2025-strategic-plan; 2022. Accessed February 11, 2025.
12. Dunbar N, Forrester M, Patrick R, Thanekar U, Ananthapavan J. Balancing economic and social cost and environmental sustainability: a case study of reusable isolation gowns. *Appl Health Econ Health Policy*. 2025;23(1):155–159.
13. Kindred M, Shabrina Z, Zakiyah N. Exploratory approach to incorporating carbon footprint in Health Technology Assessment (HTA) modelling: cost-effectiveness analysis of health interventions in the United Kingdom. *Appl Health Econ Health Policy*. 2024;22(1):49–60.
14. Husereau D, Drummond M, Augustovski F, et al. Consolidated Health Economic Evaluation Reporting Standards (CHEERS) 2022 explanation and elaboration: a report of the ISPOR Cheers II good practices task force. *Value Health*. 2022;25(1):10–31.
15. Turner RM, Dieng M, Khanna N, et al. Performance of long-term CT and PET/CT surveillance for detection of distant recurrence in patients with resected stage IIIA–D melanoma. *Ann Surg Oncol*. 2021;28(8):4561–4569.

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16. Williams JTW, Dieng M, Bell K, McAlister S, Morton RL. Healthcare costs and carbon emissions of Stage III melanoma surveillance imaging. *Appl Health Econ Health Policy*. 2026;24(1):231–241.
17. Dieng M, Turner RM, Lord SJ, et al. Cost-effectiveness of PET/CT surveillance schedules to detect distant recurrence of resected Stage III melanoma. *Int J Environ Res Public Health*. 2022;19(4):4.
18. Dummer R, Hauschild A, Santinami M, et al. Five-year analysis of adjuvant dabrafenib plus trametinib in Stage III melanoma. *N Engl J Med*. 2020;383(12):1139–1148.
19. Long GV, Swetter SM, Menzies AM, Gershenwald JE, Scolyer RA. Cutaneous melanoma. *Lancet*. 2023;402(10400):485–502.
20. Machet L, Nemeth-Normand F, Giraudeau B, et al. Is ultrasound lymph node examination superior to clinical examination in melanoma follow-up? A monocentre cohort study of 373 patients. *Br J Dermatol*. 2005;152(1):66–70.
21. Elliott TM, Whiteman DC, Olsen CM, Gordon LG. Estimated healthcare costs of melanoma in Australia over 3 years post-diagnosis. *Appl Health Econ Health Policy*. 2017;15(6):805–816.
22. Malmberg R, Loosveld JH, Schilte HP, Burdorf A, Leeuwen RWF van. Effect of alternative dosing strategies of pembrolizumab and nivolumab on healthcare emissions in the Netherlands: a carbon footprint analysis. *Lancet Planet Health*. 2024;8(11):e915–e923.
23. Dokal K, Morris M, Spooner R, Perkins P. The carbon footprint of a hospice. *BMJ Support Palliat Care*. 2024;14(e1):e496–e499.
24. Long GV, Stephen Hodi F, Lipson EJ, et al. Overall survival and response with nivolumab and Relatlimab in advanced melanoma. *NEJM Evid*. 2023;2(4):EVIDoa2200239.
25. Australian Institute of Health and Welfare. Palliative care services in Australia, Characteristics of palliative care-related hospitalisations. Australian Institute of Health and Welfare. https://www.aihw.gov.au/reports/palliative-care-services/palliative-care-services-in-australia/contents/admitted-patient-palliative-care/characteristics-palliative-care-hospitalisations; 2025. Accessed June 27, 2025.
26. McAlister S, Barratt AL, Bell KJ, McGain F. The carbon footprint of pathology testing. *Med J Aust*. 2020;212(8):377–382.
27. McAlister S, McGain F, Breth-Petersen M, et al. The carbon footprint of hospital diagnostic imaging in Australia. *Lancet Reg Health West Pac*. 2022;24:100459.
28. Williams JTW, Bell K, Morton RL, Guitera P, Martin L, McAlister S. The climate change impact of surgical and non-surgical skin cancer appointments: a process-based life cycle assessment. *Open Science Framework*. 2025. Forthcoming.
29. Dieng M, Lord SJ, Turner RM, et al. The impact of surveillance imaging frequency on the detection of distant disease for patients with resected Stage III melanoma. *Ann Surg Oncol*. 2022;29(5):2871–2881.
30. Australian Institute of Health and Welfare. Health expenditure Australia 2022-23. Australian Institute of Health and Welfare. https://www.aihw.gov.au/reports/health-welfare-expenditure/health-expenditure-australia-2022-23/; 2024. Accessed April 23, 2025.
31. Australian Government Department of Health. Disability and ageing. Guidelines for preparing submissions to the Pharmaceutical Benefits Advisory Committee. version 5.0. Pharmaceutical Benefits Advisory Committee Guidelines. https://pbac.pbs.gov.au; 2016. Accessed August 8, 2025.
32. Eggermont AMM, Chiarion-Sileni V, Grob JJ, et al. Prolonged survival in Stage III melanoma with ipilimumab adjuvant therapy. *N Engl J Med*. 2016;375(19):1845–1855.
33. Blank CU, Lucas MW, Scolyer RA, et al. Neoadjuvant nivolumab and ipilimumab in resectable Stage III melanoma. *N Engl J Med*. 2024;391(18):1696–1708.
34. Smith AJB, Tennison I, Roberts I, Cairns J, Free C. The carbon footprint of behavioural support services for smoking cessation. *Tob Control*. 2013;22(5):302–307.
35. McAlister S, Morton RL, Barratt A. Incorporating carbon into health care: adding carbon emissions to health technology assessments. *Lancet Planet Health*. 2022;6(12):e993–e999.
36. Aldy JE, Kotchen MJ, Stavins RN, Stock JH. Keep climate policy focused on the social cost of carbon. *Science*. 2021;373(6557):850–852.
37. NSW Government. Carbon emissions in the investment framework. NSW Treasury. https://www.nsw.gov.au/departments-and-agencies/nsw-treasury/documents-library/tpg24-34; 2024. Accessed August 11, 2025.
38. Rennert K, Errickson F, Prest BC, et al. Comprehensive evidence implies a higher social cost of CO2. *Nature*. 2022;610(7933):687–692.
39. Ducrot C, Péron J, Delaye M, Pérol D, Durand-Zaleski I, Piffoux M. Integrating environmental impact in health technology assessment: an exploratory study. *Pharmacoeconomics*. 2025;43(10):1205–1222.
40. Bressler RD. The mortality cost of carbon. *Nat Commun*. 2021;12(1):4467.
41. Polisena J, De Angelis G, Kaunelis D, Gutierrez-Ibarluzea I. Environmental impact assessment of a health technology: a scoping review. *Int J Technol Assess Health Care*. 2018;34(3):317–326.
42. Pinho-Gomes AC, Yoo SH, Allen A, Maiden H, Shah K, Toolan M. Incorporating environmental and sustainability considerations into health technology assessment and clinical and public health guidelines: a scoping review. *Int J Technol Assess Health Care*. 2022;38(1):e84.
43. Guirado-Fuentes C, Abt-Sacks A, Trujillo-Martín M del M, et al. Main challenges of incorporating environmental impacts in the economic evaluation of health technology assessment: a scoping review. *Int J Environ Res Public Health*. 2023;20(6):6.
44. Antonazzo IC, Gribaudo G, Ye L, et al. Health technology assessment and environmental impact: a scoping review of state of art and future perspective. *Appl Health Econ Health Policy*. 2025;23(6):1027–1038.
45. National Institute for Health and Care Excellence (NICE). NICE listens: public dialogue on environmental sustainability. National Institute for Health and Care Excellence (NICE). https://www.nice.org.uk/about/what-we-do/our-research-work/nice-listens; 2023. Accessed February 10, 2025.
46. Pugliesi PS, Frick H, Guillot S, et al. Cost of carbon in the total cost of a healthcare procedure: example of micro-costing study in a French setting. *Appl Health Econ Health Policy*. 2025;23(2):265–275.
47. de Preux L, Rizmie D. Beyond financial efficiency to support environmental sustainability in economic evaluations. *Future Healthc J*. 2018;5(2):103–107.
48. Moore FC, Drupp MA, Rising J, Dietz S, Rudik I, Wagner G. Synthesis of evidence yields high social cost of carbon due to structural model variation and uncertainties. *Proc Natl Acad Sci U S A*. 2024;121(52):e2410733121.
49. Watts N, Andrew T, Bagenal J, et al. MedZero: carbon analytics for health care, by health care, at scale. *Lancet*. 2025;407(10525):205–207.
50. Piffoux M, Le Tellier A, Taillemite Z, Ducrot C, Taillemite S. Carbon footprint of oral medicines using hybrid life cycle assessment. *J Cleaner Prod*. 2024;475:143576.
51. HealthcareLCA. A database of healthcare's environmental impacts. Centre for Planetary Health and Sustainable Care. https://healthcarelca.com/; 2025. Accessed March 20, 2025.
52. McAlister S, Barratt A, Bell K, McGain F. How many carbon emissions are saved by doing one less MRI? *Lancet Planet Health*. 2024;8(6):e350.
53. Marsh K, Ganz ML, Hsu J, Strandberg-Larsen M, Gonzalez RP, Lund N. Expanding health technology assessments to include effects on the environment. *Value Health*. 2016;19(2):249–254.
54. Hensher M. Incorporating environmental impacts into the economic evaluation of health care systems: perspectives from ecological economics. *Resour Conserv Recycl*. 2020;154:104623.
55. Kingma SL, Van Bree EM, Rutten-Van Mölken MPMH, IJzerman MJ, Bree EM van, Mölken MPMHR van, IJzerman MJ. Exploring methods to include carbon emissions into an HTA: the case of remote patient monitoring. *Value Health*. 2025;0(0).
56. Oldenburg J, Keil M, Maaß L, Lange O, Rogowski W. Study protocol for a Delphi process to develop a climate impact extension to the Consolidated Health Economic Evaluation Reporting Standard (CHEERS) 2022—the CHEERS ClimatE checklist. *BMJ Open*. 2025;15(9):e094617.